Abstract

Alzheimer's Disease (AD), currently the most common cause of dementia, is characterized by the accumulation of amyloid plaques and neurofibrillary tangles in the brain, leading to neuronal degeneration and brain atrophy. Patients with AD experience progressive cognitive and behavioral impairments, including memory loss, decreased concentration, impaired reasoning, and difficulties with daily functioning. Globally, over 55 million people are living with dementia, with AD accounting for 60-70% of cases. This places a substantial burden on caregivers, who often experience emotional stress, depression, anxiety, and a reduced quality of life (QoL). To address this strain, this study explores the development of an assistive ecosystem for Alzheimer's care: Adaptive Learning Zone Using Artificial Intelligence (ALZUAI). ALZUAI integrates voice assistants, motion sensors, and environmental cues within a privacy-preserving architecture, providing active notifications (real-time task reminders, safety alerts, and gentle guidance) to users, all without requiring wearable devices within the home, a known barrier in AD care. Many existing smart-home dementia studies focus on issue detection rather than active assistance, a gap ALZUAI addresses through proactive contextual prompts. Machine learning personalizes reminders and notifications based on behavioral trends, and caregivers receive updates via SMS or mobile application. By offering continuous, unobtrusive monitoring and proactive support, ALZUAI aims to reduce caregiver oversight burden, enhance patient safety and improve overall QoL for both patient and caregiver.

Introduction

Approximately 7.4 million Americans aged 65 and older are living with Alzheimer's Disease (AD), typically surviving four to eight years post-diagnosis. Because AD demands highly individualized support, care is largely provided by unpaid family members and friends within the community. This care requires immense flexibility, particularly when navigating the drastic behavioral changes common to AD: wandering, aggression, or inappropriate sexual behavior. Managing these unpredictable environments places a severe psychological and physical toll on caregivers. Their daily responsibilities span a wide and exhausting range, from basic tasks like grocery shopping and bill paying to intensive hands-on care like bathing and medication management (CDC and Alzheimer's Association). Consequently, despite a recent increase in available caregiver resources, full-time caregivers experience a noticeable reduction in their overall Quality of Life (QoL). The chronic stress of Alzheimer's caregiving frequently manifests as social isolation, loneliness, compromised immune function, and a higher reliance on psychoactive medications.

Historically, several measures have been explored in order to bridge the gap between effective AD care and caregiver QoL. The introduction of smart home technologies, such as ambient and motion sensors, enable continuous, real-time tracking of behaviors like sleep patterns and gait to assist Alzheimer's patients aging in place. While these technologies improve safety and slightly reduce caregiver stress, their effectiveness is hindered by technical unreliability, ethical concerns regarding surveillance, and a lack of standardized evaluation frameworks (Shaik et al., 2025). To combat existing hurdles in current research, a shift from simple issue detection to proactive assistance is needed. ALZUAI bridges the gap between passive monitoring and active care by introducing a low-cost, multi-modal environmental framework that delivers personalized, real-time guidance without compromising user privacy and data autonomy.

Literature Review

Previous research regarding the creation of adaptive ecosystems has proven the concept effective, however, there still remains a need for active assistance in the form of dynamic notifications. Qiao et al. conducted a study that investigates the relationship between the adoption of Assistive Smart Home Technology (SHT) and the Health-Related Quality of Life (HRQoL) among older adults, with a specific focus on how depressive symptoms moderate this relationship. Through the use of cross-sectional surveys, their research concluded that using assistive smart home technologies is directly associated with a significant positive increase in an older adult's health-related quality of life. By introducing depressive symptoms as a moderator, the study proves a vital behavioral concept; vulnerable individuals experience a high marginal utility from automated environments. When a person is experiencing executive dysfunction due to depression or early cognitive decline, the smart home minimizes the friction of daily life (e.g., remembering tasks, manual labor, environment adjustments). However, the paper is fundamentally limited by its reliance on basic consumer hardware metrics, failing to evaluate the deeper data pipelines or active notification frequencies required for true clinical intervention.

Martins et al.'s study describes a Smart Home Orchestration Framework designed to transition traditional smart homes from static, "trigger-action" systems (e.g., if motion detected, then turn on light) to proactive, intelligent ecosystems powered by Artificial Intelligence (AI). The framework relies on a multi-module architecture:The Core Module, The Smart Module,The Visualization/Mobile Module. The study addresses an existing automation cognitive gap. Standard commercial smart homes depend on manual rule creation, which experiences a high rate of user fatigue and fails to adapt to fluid, real-world routine changes. By integrating an LSTM network within the Smart Module, the framework shifts the smart environment from reactive to predictive. The network doesn't just evaluate isolated inputs; it processes sequences of behaviors (e.g., adjusting light intensity relative to room occupancy history and air conditioning modes). To ensure user agency and reduce automation fatigue, the framework delivers predictive routine recommendations to a mobile interface via push notifications, allowing users to explicitly accept or reject new automations while providing transparent dashboards tracking energy savings and carbon footprint metrics. While the framework implements manual overrides via the mobile UI, the study provides limited outlier evaluation on how the LSTM handles rapid, erratic deviations in user behavior without muddying long-term recommendation accuracy.

Jiang, D. proposes an AI-driven framework for the aging-friendly renovation of smart homes. Rather than relying on static configurations, the framework dynamically captures multi-dimensional behavioral data from elderly residents to generate and iterate personalized, adaptive spatial layouts. Jiang's multi-Model AI framework combines various AI methodologies, Long Short-Term Memory (LSTM), Reinforcement Learning (RL), and Conditional Generative Adversarial Network (cGAN), to successfully build a closed-loop “perception-cognition-response” pipeline. His research highlighted improvements in key areas such as fall risk, with scores in high-hazard zones (kitchens and bathrooms) reduced by 50%, and security. To safeguard sensitive user data, the system processes raw sensor inputs locally on edge computing devices and integrates blockchain technology for secure, un-tamperable data access logging. The system's primary response is to independently create and optimize physical interior design solutions, i.e., generating visual spatial layout drafts, that adhere to existing structural constraints. Despite the effectiveness of this framework in relation to design, it fails to include an accurate, fully functional notification system.

Research conducted by Shaik, et al., proposes an integrated remote monitoring framework which makes use of various remote sensing technologies designed for patients with Alzheimer’s Disease and Related Dementia (ADRD). Following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework, the authors conducted a rigorous systematic literature search across four databases (Google Scholar, PubMed, IEEE Xplore, and DBLP) with focus on four thematic clusters: Existing remote monitoring technologies, Balancing practicality and empathy in eldercare, Security and Privacy protocols within remote patient monitoring systems, and User-centered design paradigms tailored for cognitive decline. This framework identified a major gap in the creation of an adaptive ecosystem and the individuals it was meant to serve, with mention of an existing disproportionate focus on technical data acquisition relative to pragmatic application. Engineers are successfully creating high-tech tracking systems, but clinicians and caregivers are left with systems that lack rigorous evaluation regarding human factors. Shaik et al.’s findings underscores that while adaptive environments can react physically, they still desperately need optimized, user-centric notification systems to bridge the gap between automated home adjustments and human caregiver awareness.

Experiment

[Placeholder]

Conclusion

[Placeholder]

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